Characterisation of Children's Head Motion for Magnetic Resonance Imaging With and Without General Anesthesia

ISMRM MoCo workshop 2022

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KØBENHAVNS UNIVERSITET

Declarations with respect to the subject matter of this presentation

The presented work on characterization of children's motion patterns was

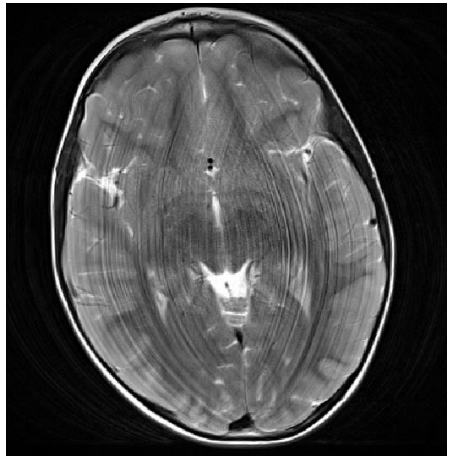
- funded by the Elsass foundation, Danish Childhood Cancer Foundation and the Novo Nordisk and Novozymes Talent Program

- carried out in cooperation with TracInnovation Aps who provided the motion tracking data for the pediatric population. AE and JS, as well as MG's husband are employees at TracInnovations, Ballerup, Denmark.

- ongoing collaboration with TracInnovation Aps in a prospective clinical study on the effectiveness of preparation and motion correction in pediatric patients (Danish national ethics protocol H-20055983)

Introduction

MRI artefacts and their consequences



- patient head motion due to long acquisition times
  [Afacan et al. Pediatr Radiol. 2016]
- artefacts manifest as ghosting, blurring or signal variations, overall image quality and resulting in unsuccessful diagnoses
  [Godenschweger et al. PhysMed Biol. 2016]

- 20% percentage of at least partly repeated MRI examinations, estimated increased cost of 115,000 US dollars/scanner/year in the US due to motion
  [Andre et al. J Am Coll Radiol. 2015]

Introduction
Clinical MRI of children

- tend to move more in the MR scanner than adults *[De Bie et al. Eur J Pediatr. 2010]*
  - more active, lie less still
  - intimitated by scanner size
  - exposure to loud noises

- Diagnostic MRI of newborns, infants, and toddlers is typically conducted using sedation or GA *[Artunduaga et al. Pediatr. Radiol. 2021]*
  - 61–100% for children between 4 and 6 years *[Runge et al. Eur J Radiol. (2018)]*
  - 80% for children aged 1-6 years *[Jaimes et al. Am J Roentgenol. 2021]*
  - 40% for children aged 7–12 years *[Jaimes et al. Am J Roentgenol. 2021]*

Introduction
Consequences of anesthesia

- risk of sedation-related adverse events *[Cravero et al. Anesthesia & Analgesia 2009]*
- neurotoxic effects, particularly when administered to young children *[Stratmann et al. Anesthesia & Analgesia 2011]*
- FDA label changes for common general anesthetics and sedative agents in young children in 2017 *[Andropoulos et al. N Engl J Med 2017]*

- increases the patient's waiting times as well as the costs of the scan
  - additional cost for using GA in MRI estimated to 319,000 US dollars/scanner/year in Denmark *[Slipsager et al. J Magnet Reson. Imaging. (2020)]*

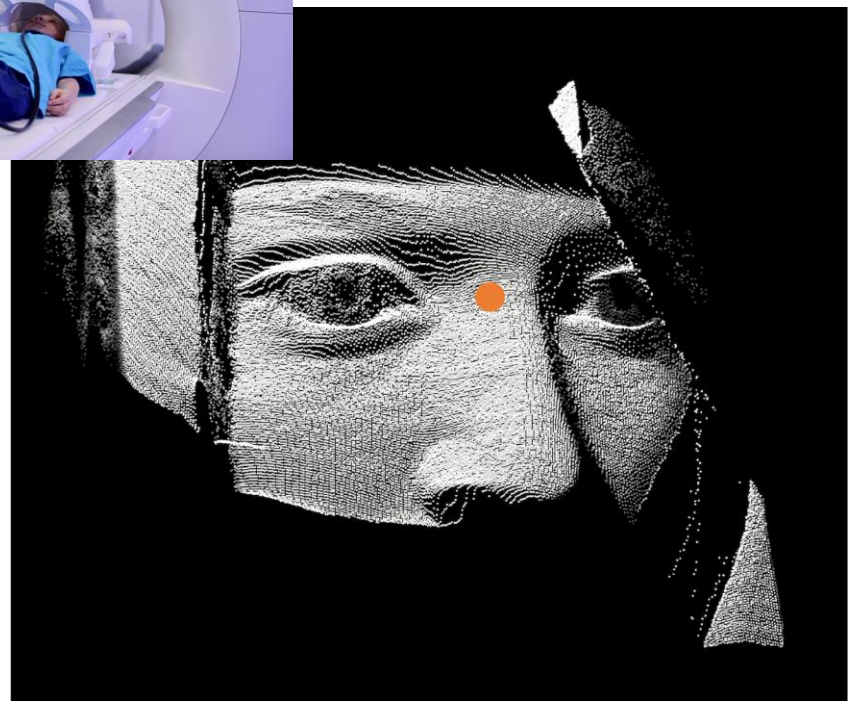
Retrospective study about children's head motion patterns during MRI

- Analyze motion patterns of children in order to draw possible conclusions about predominant movement habits

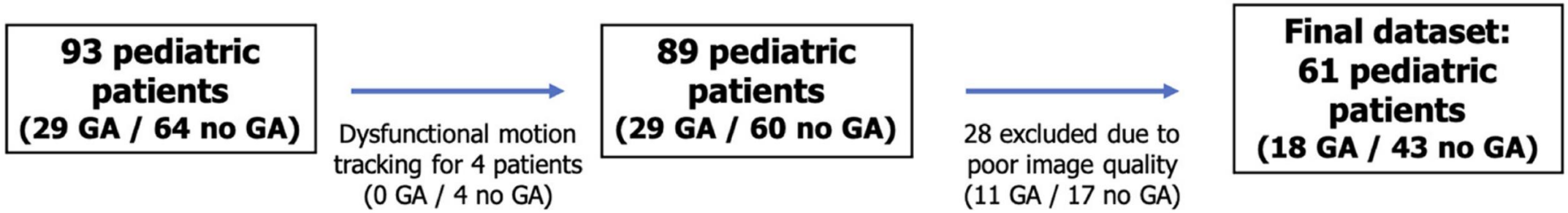
Characterisation of Children's Head Motion for Magnetic Resonance Imaging With and Without General Anaesthesia

Hannah Eichhorn[1,2], Andreea-Veronica Vascan[1,3], Martin Nørgaard[1,4], Andreas H. Ellegaard[5], Jakob M. Slipsager[5,6,7], Sune Høgild Keller[7], Lisbeth Marner[7,8] and Melanie Ganz[1,3*]

[Eichhorn et al. Frontiers in Radiology 2021]



Methods
Study Population



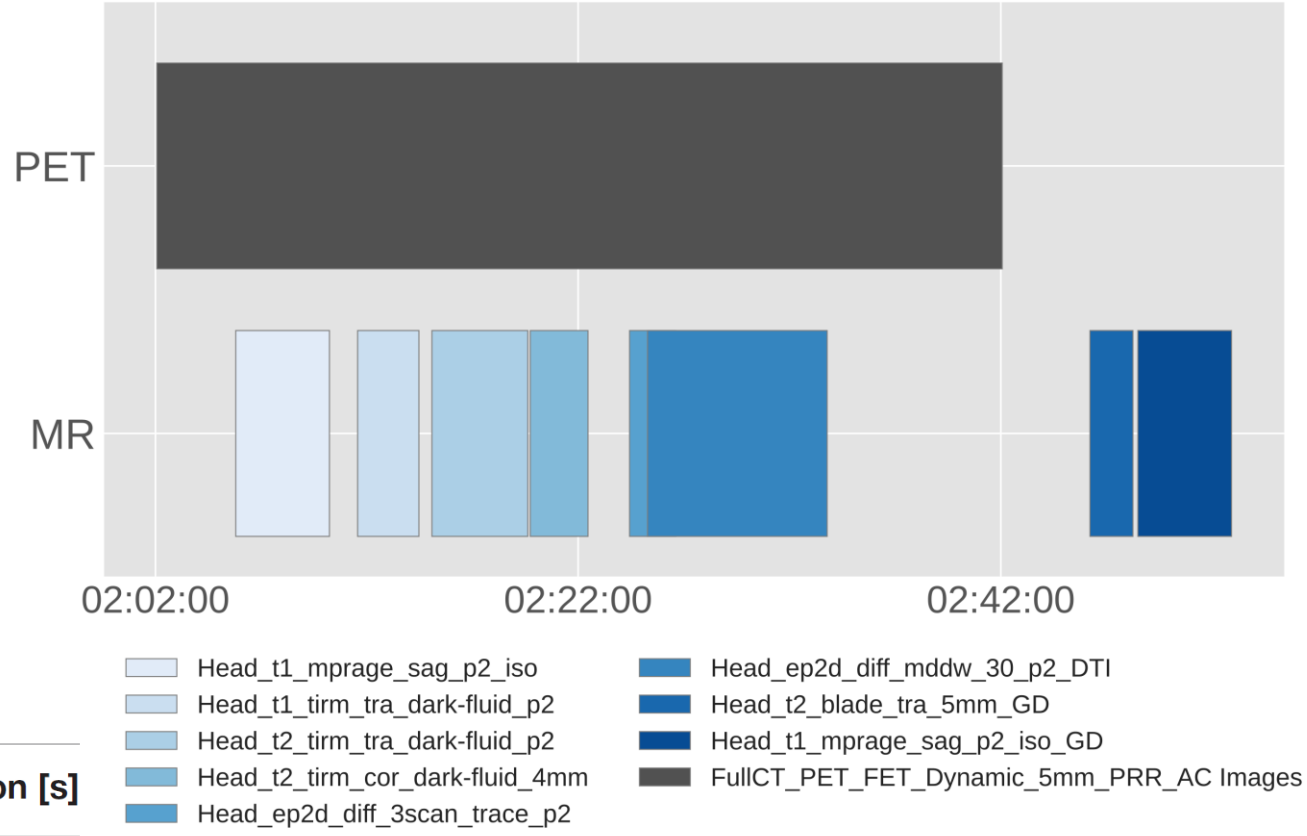
- Diagnostic accuracy and clinical impact of [^{18}F]FET PET in childhood CNS tumors
- Danish National Committee on Health Research Ethics (approval H-6-2014-095)
- clinicaltrials.gov (NCT03402425)

	GA	No GA	All
Number of patients (sex)	18 (10 male)	43 (28 male)	61 (38 male)
Mean age ± std	5.94 ± 2.75	12.65 ± 3.04	10.67 ±4.25
Age range	1–13	7–19	1–19

Methods

Scanner setup

- mMR Biograph hybrid PET/MRI scanner (Siemens Healthineers, Erlangen,Germany)
- April 2015 and January 2019
- child brain tumor protocol



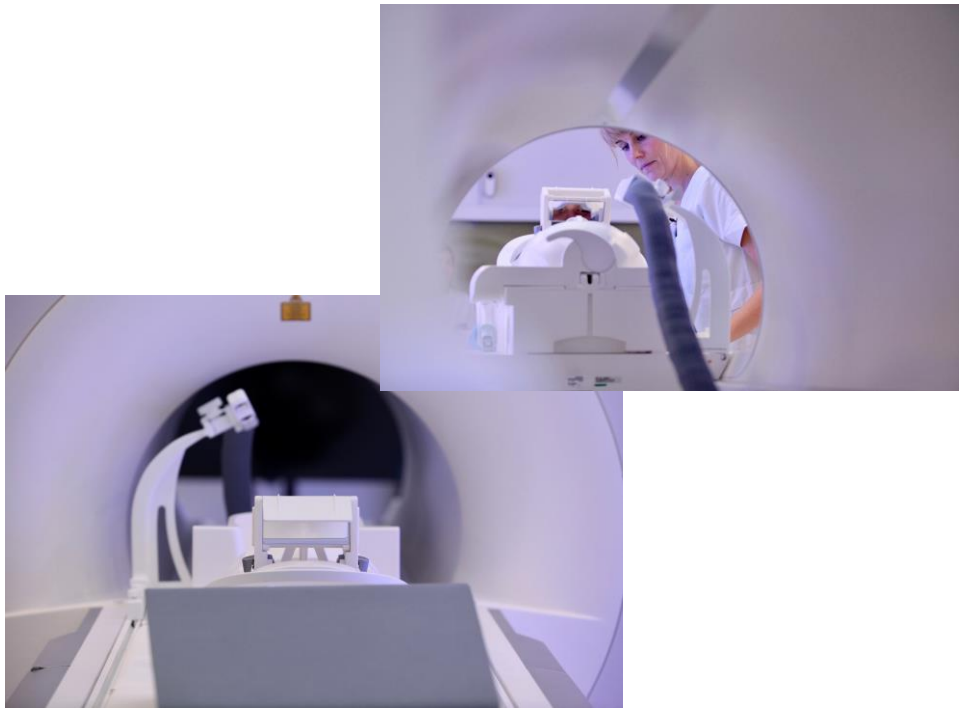
Sequence	Scan duration [s]
T_1-weighted MPRAGE	266
T_1-weighted STIR	174
T_2-weighted FLAIR (transversal)	272
T_2-weighted FLAIR (coronal)	164
T_2-weighted Blade	122

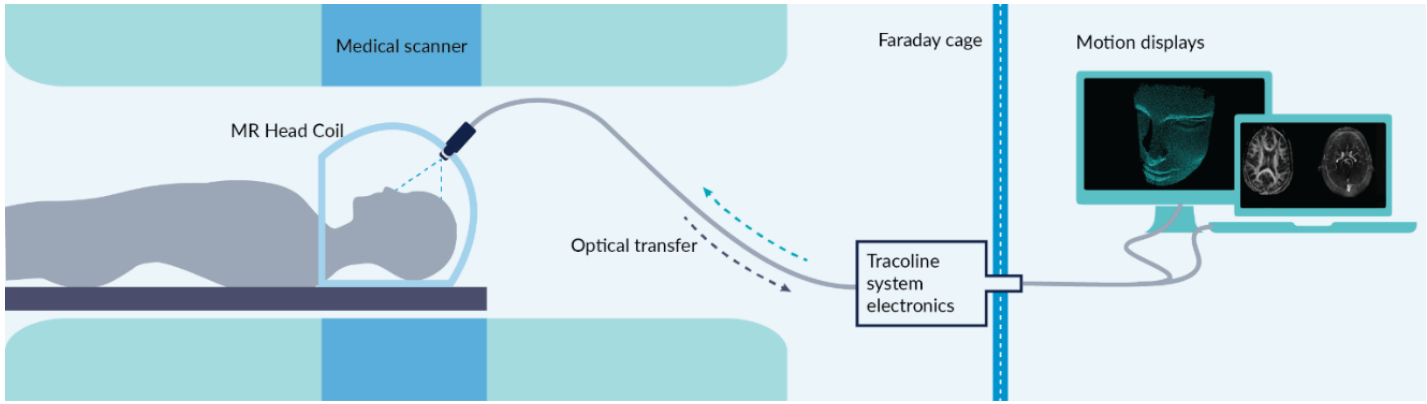
Note that the protocol consisted of both MR and PET scans, which is why the median wall time from the first to the last MR sequence was longer than the sum of the scan durations.

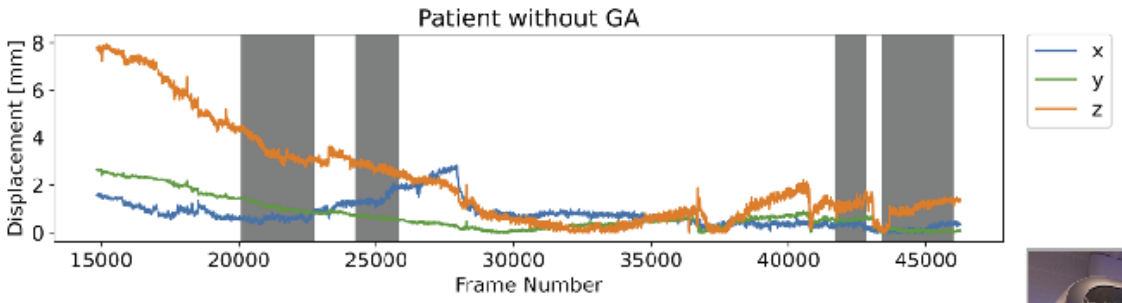
Methods

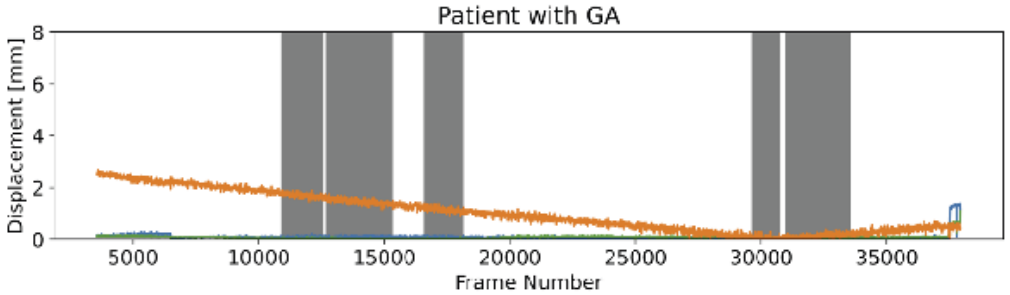
Motion tracking

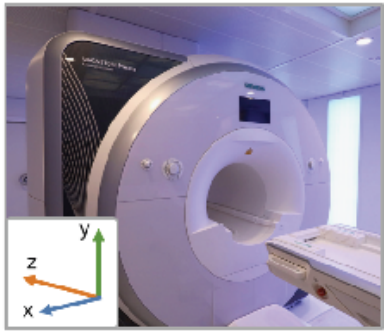
- Tracoline system





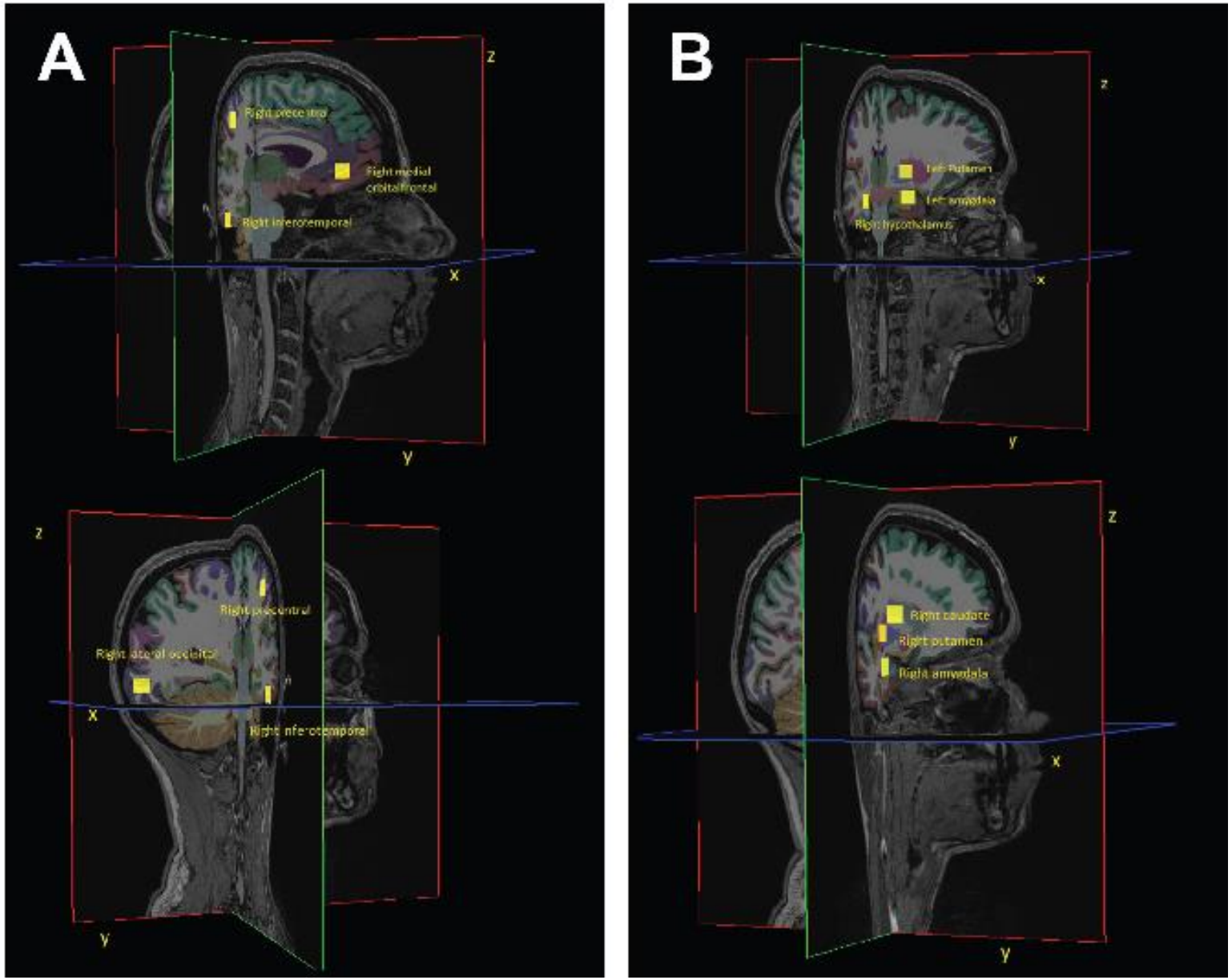






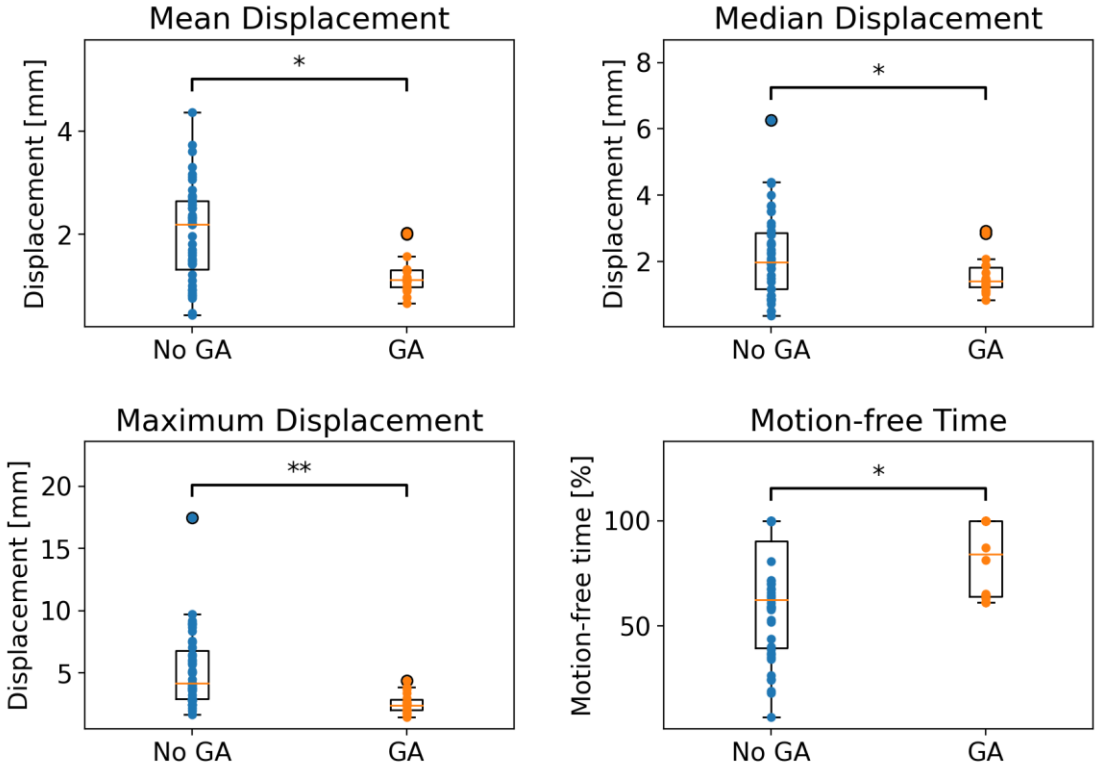
Methods
Regions of Interest

- 4 cortical (A) and 4 subcortical (B) regions together with their centroid (yellow square)
- Only half of regions shown here



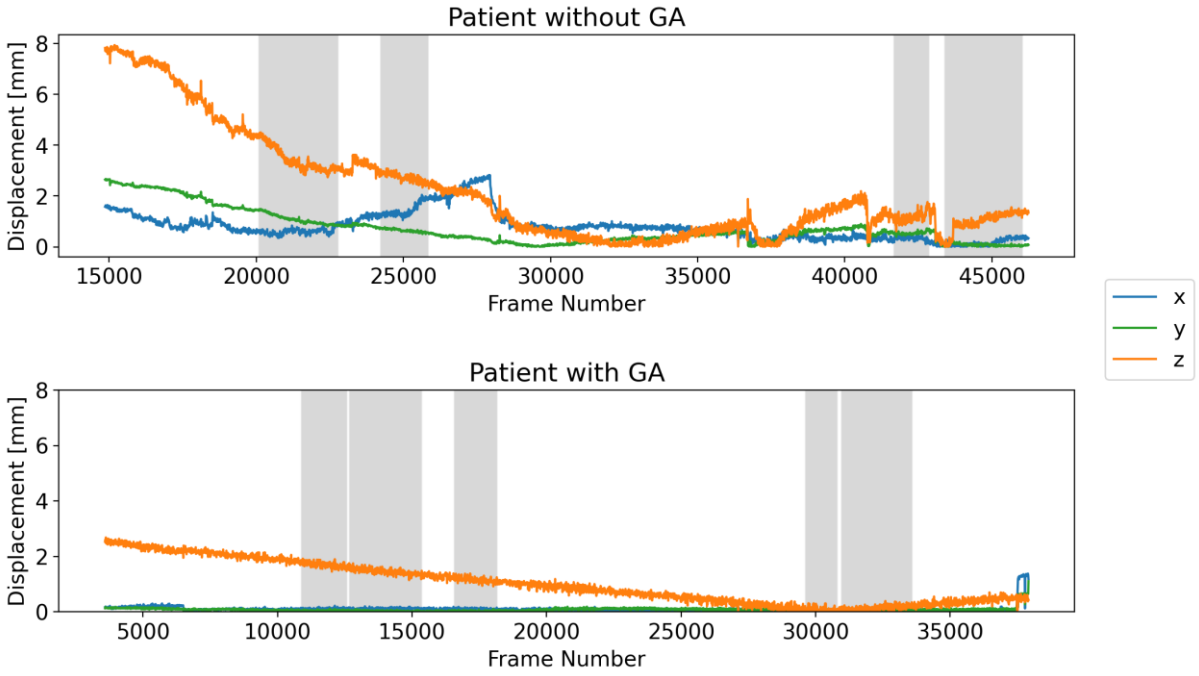
Results

Comparison of motion metrics



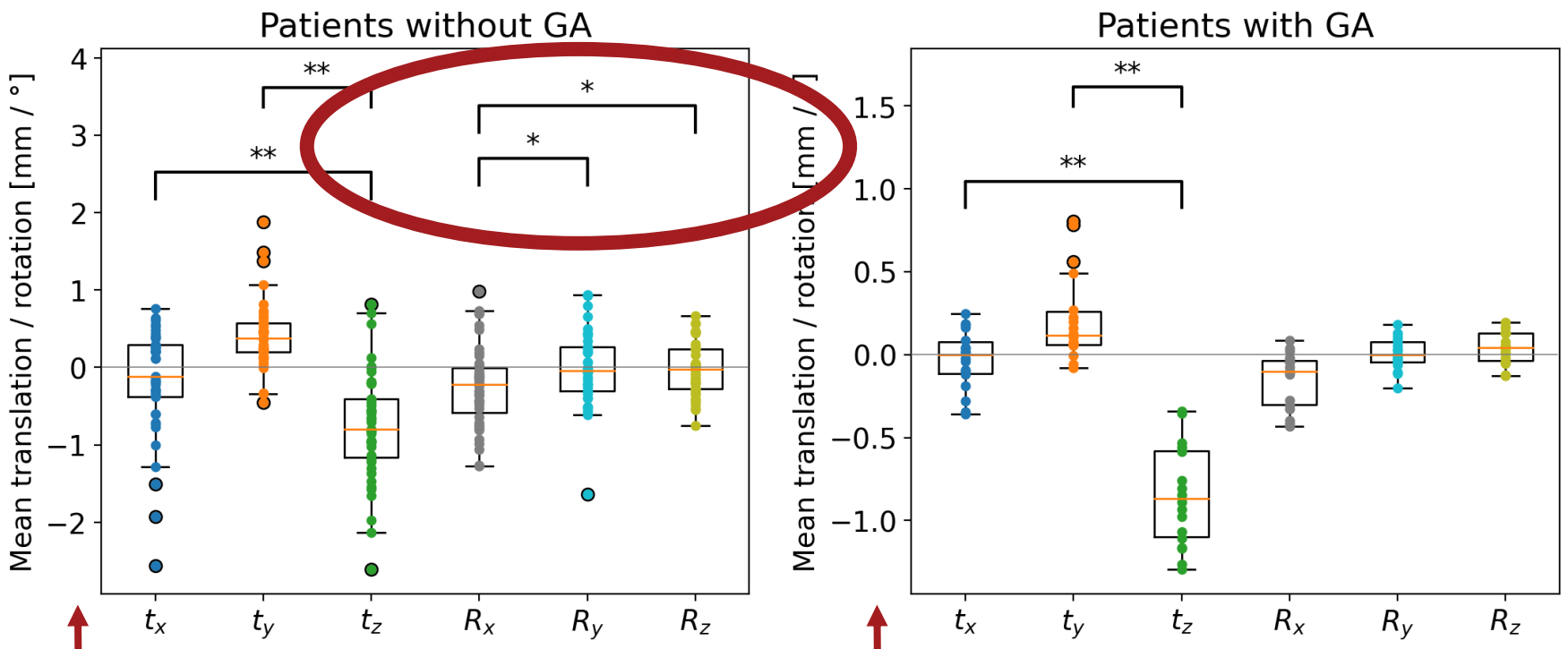
*** : p < 0.05**

**** : p < 0.001**



Results II

Decomposition of transformation

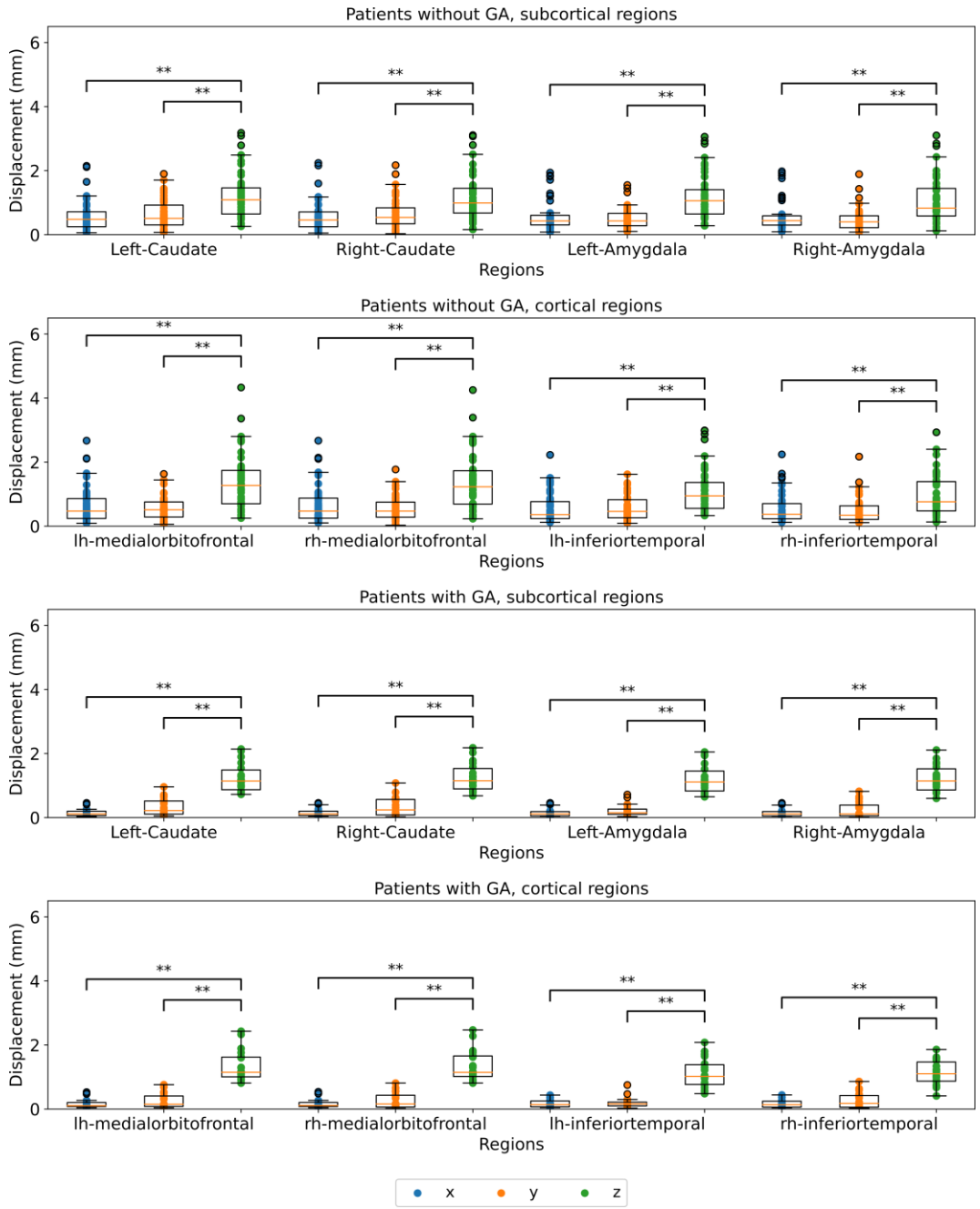


*** :** **p < 0.05**

**** :** **p < 0.001**

Results II
Regionwise Decomposition

- Difference between no GA (top) and GA bottom in size
- but no difference in size of motion between regions

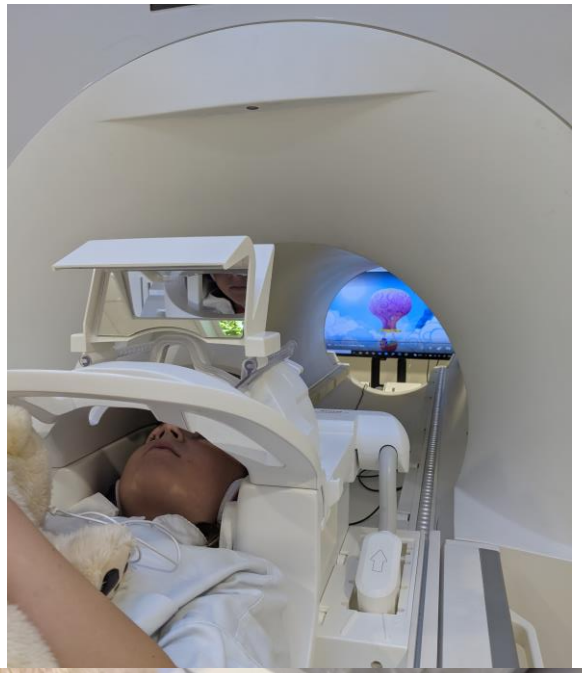


*** : p < 0.05**
**** : p < 0.001**

Discussion
Main findings

- children without GA moved significantly more than children with GA for all analyzed metrics
  - residual movement was still observed for anesthetized children

- wrt motion patterns
  - both patient groups significantly larger absolute translation along the z-axis compared to x- and y-axes -> **drift motion**
  - In children without GA, we additionally observed significantly larger rotations around the x-axis compared to the y- and z-axes -> **nodding motion**



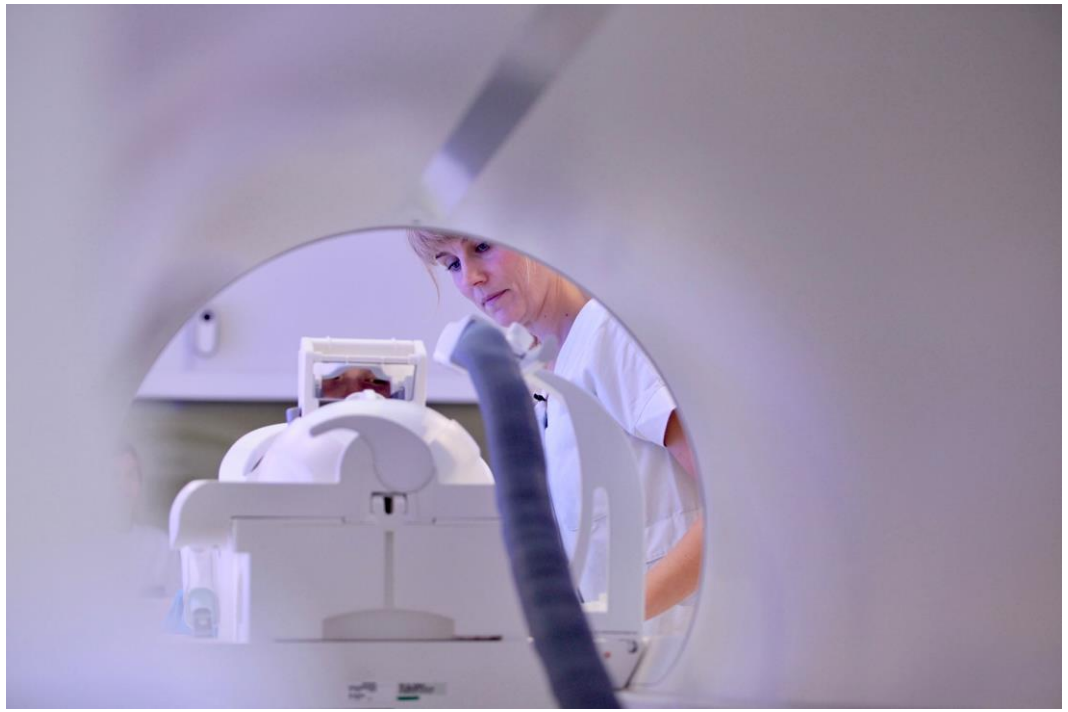
Conclusion
How should we go from here?

- How can we scan children without sedation/GA?
  - Preparation!
  - Monitoring of motion
  - Strategies to perform magnetic resonance imaging in infants and young children without sedation *[Harrington et al. Ped. Rad. 2022]*

- Can we restrict/discourage this type of motion somehow?
  - Make it less attractive to move -> **distraction by television**
  - Give direct feedback -> **training includes use of motion tracking**

Ongoing work
Neuroimaging of children without sedation can become reality

- PI: Melanie Ganz-Benjaminsen
- Located at: NRU, Rigshospitalet
- Dates: 1. April 2019 – end of 2023
- Budget: ca. 3.9 mio DKK
- Funded by Elsass Fonden



MoCo project

Purpose

- Our aim is that the majority of children referred to MR examination should be able to be scanned without sedation / GA:

- We would like to achieve this aim in two ways:
  - by preparing the child with a mobile phone app and a practice session in a practice scanner (mock scanner)

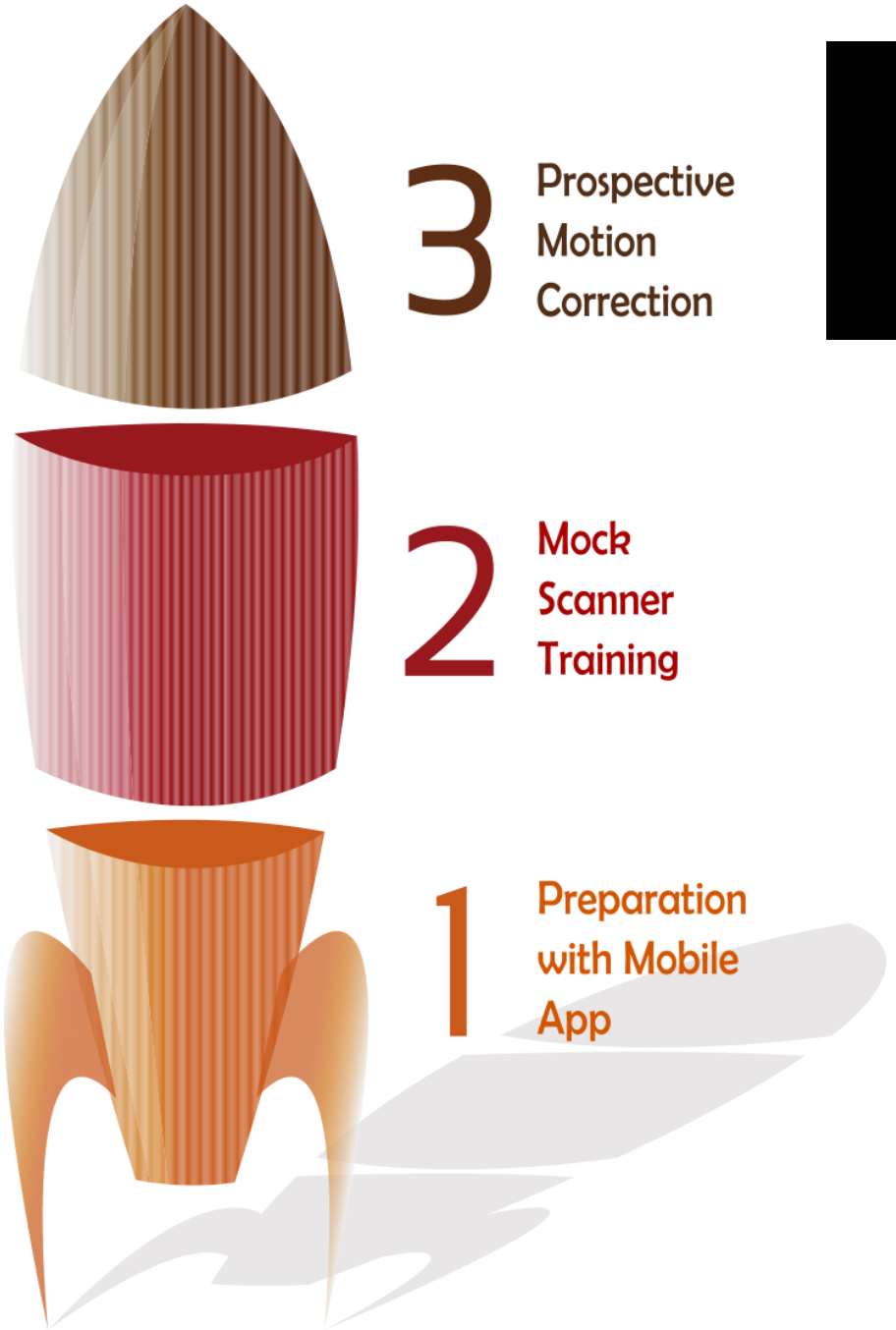
  - by correcting the images using an external motion tracking device and applying prospective motion correction during image acquisition

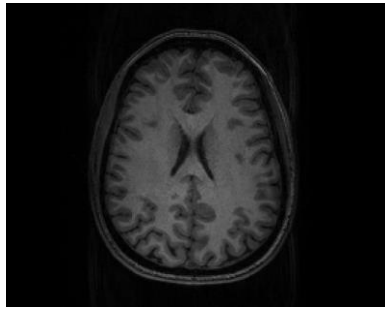
MoCo project
In-/Exclusion criteria

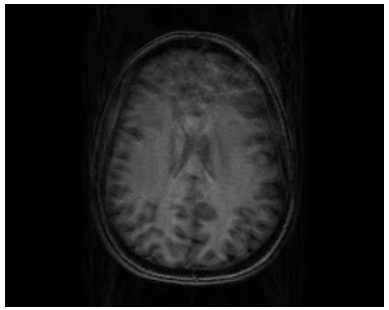
- Inclusion:
  - Age 0 -12 months or 4 -10 years
  - referred for an elective scheduled clinical cerebral MRI scan in general anesthesia

- Exclusion:
  - unstable medical condition interfering with safety of MRI scan
  - non-fluent in Danish or pronounced visual or auditory impairments
  - no major physical or developmental challenges that interfere with participation in the experimental protocol
  - contraindications for MRI (metal implants, etc.).

MoCo project
3-stage rocket





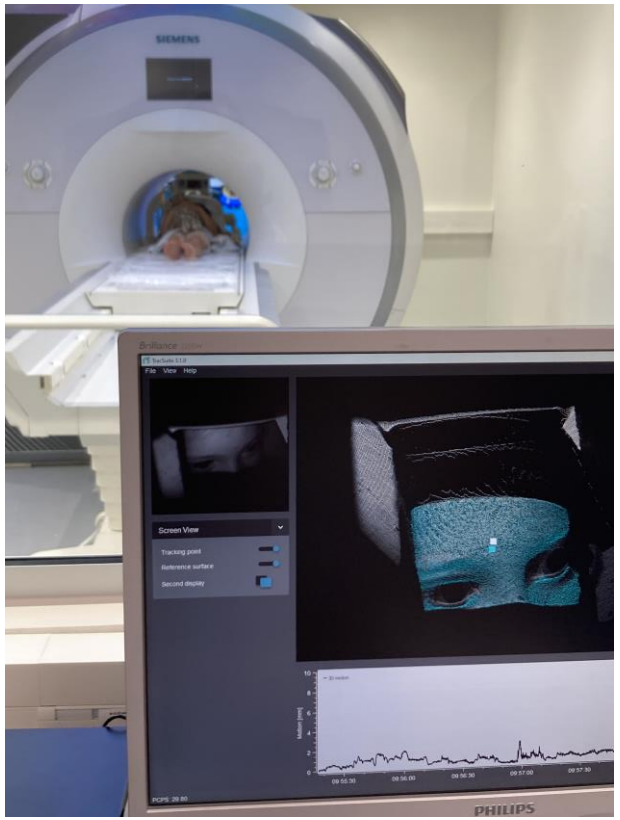


MoCo project
NRU mock scanner

Details:

- Siemens Magnetom Prisma shell
- 64- channel head coil
- Noise simulation
- Head motion tracking

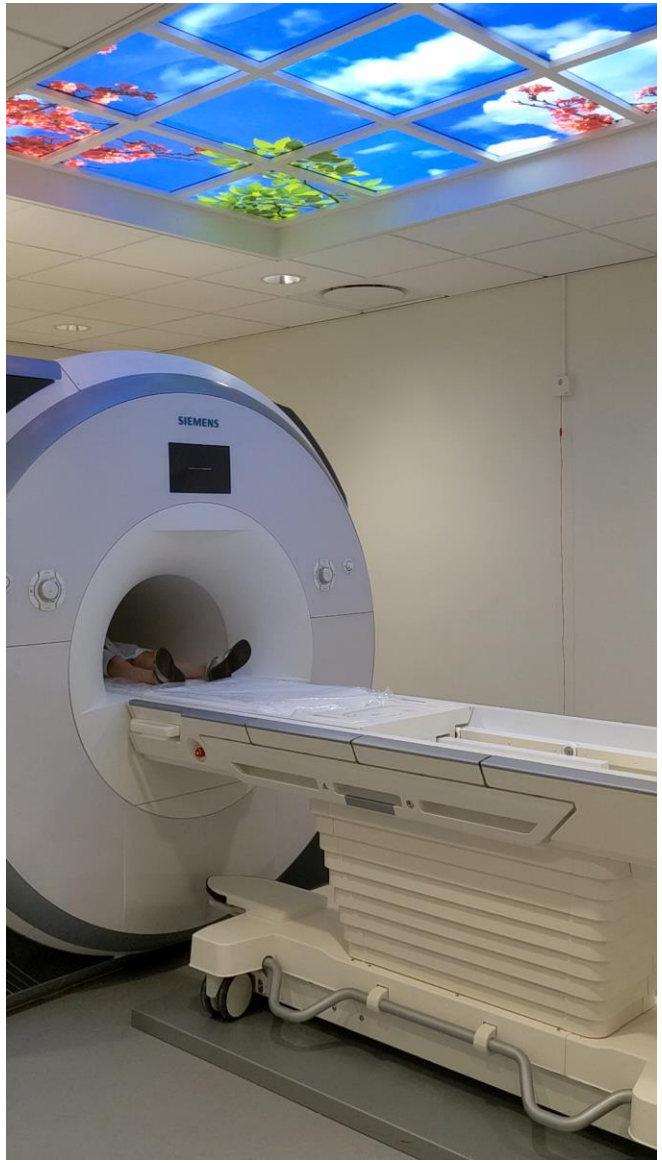
MoCo project
NRU mock scanner



Operator room setup:

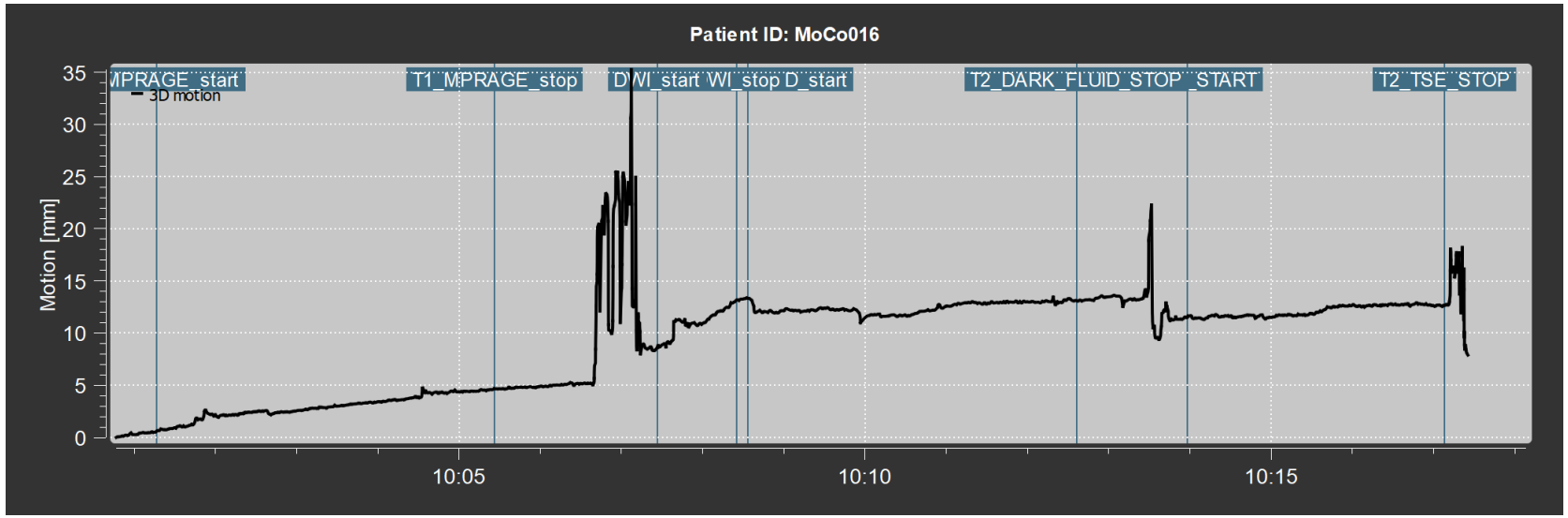
- PCs
- “fake” scan
- Sound control
- Intercom
- Movie interface

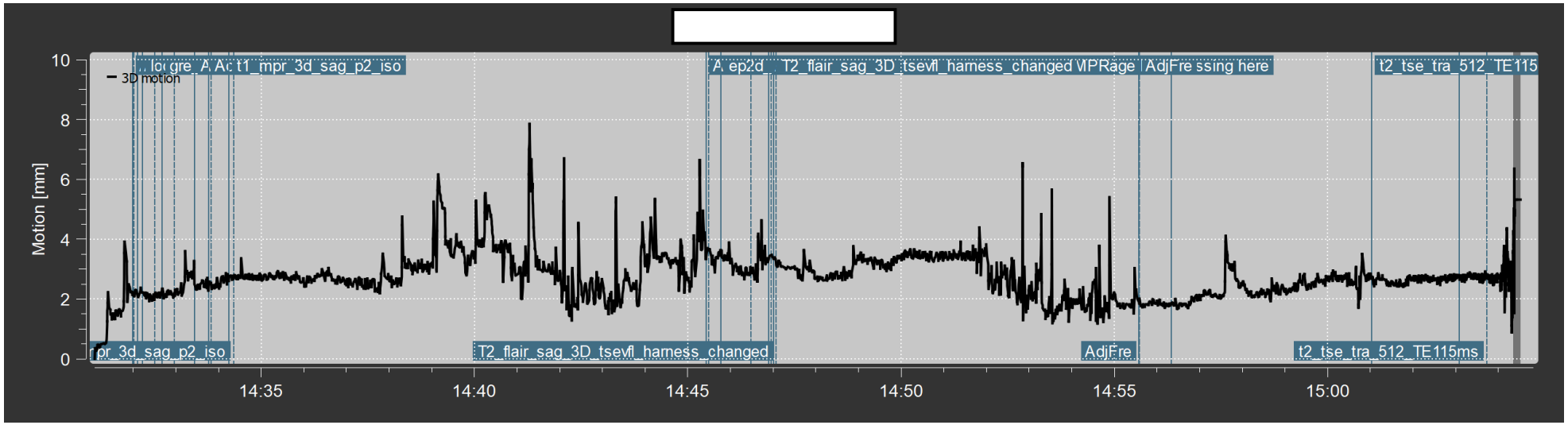
MoCo project NRU mock scanner in action



MoCo project

Motion monitoring/correction with Tracoline (in both





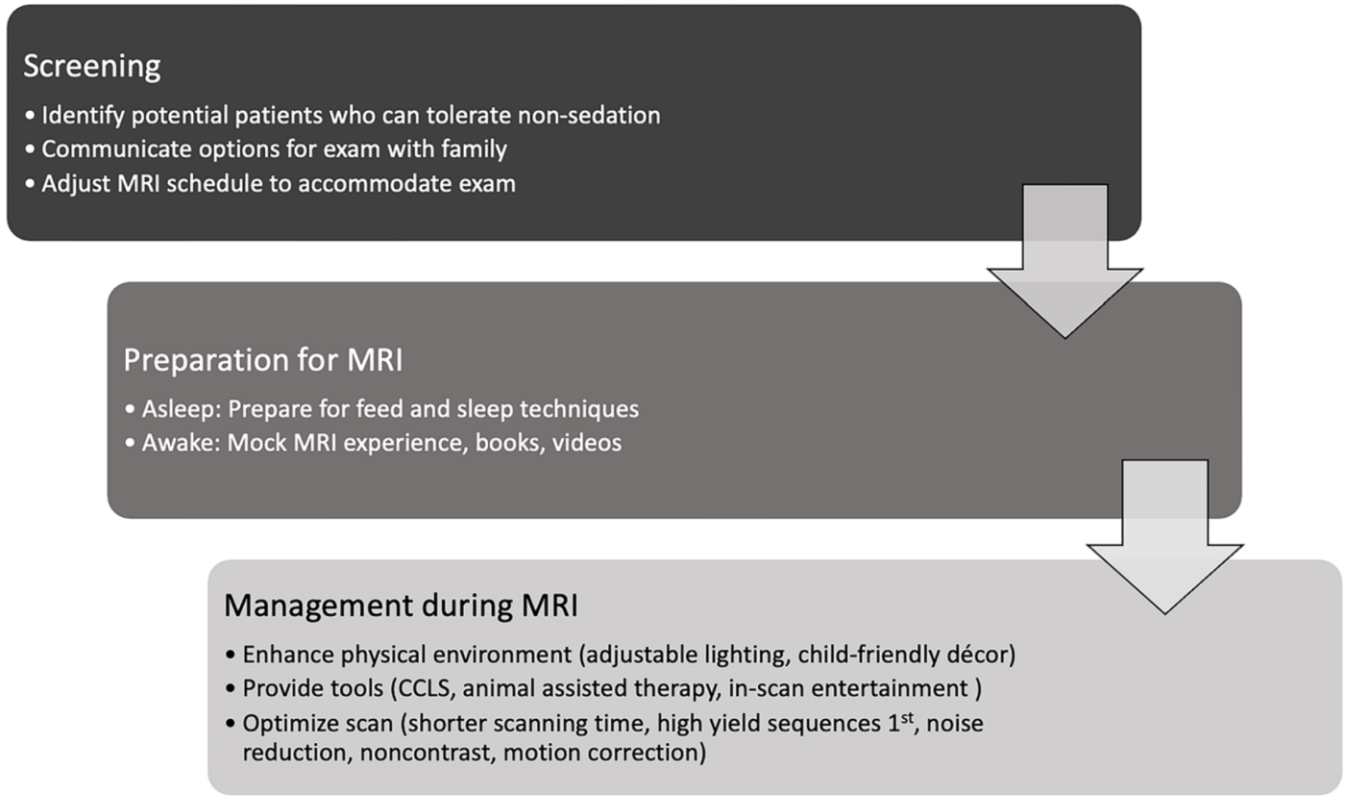
MoCo project
Preliminary results

- 25 children enrolled -> 18 have completed the full program -> 16 had a fully diagnostic scan without GA
  - 1 participant moved to GA – non-verbal autism
  - 1 participant after study offered same treatment and then successful

- No issues with entering the scanner, residual motion is very low
  - training the situation (mobile phone app) and personal interaction even more important than technical solutions?

MoCo project
Discussion

- focus on the complete clinical workflow, not only on the sequences
  [Harrington et al. Ped. Rad. 2022]



MoCo project
Future endeavours

- 50 children between 4 and 10 years by spring 2023
- moving on to children between 0 and 1 year in fall 2022, start recruitment spring 2023

- examine effectiveness of training vs. prospective/retrospective motion correction once all data is collected

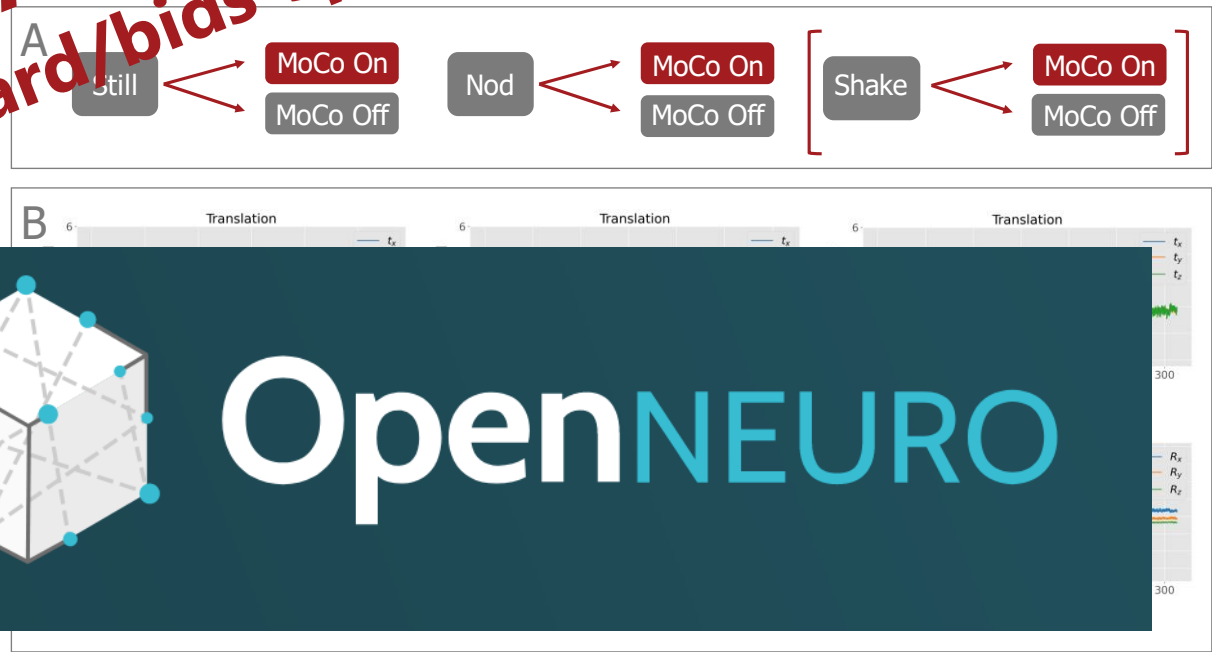
- Move on with data sharing!
  - BIDS for motion experiments
  - motion tracking database

Technical contribution

MR study about motion correction

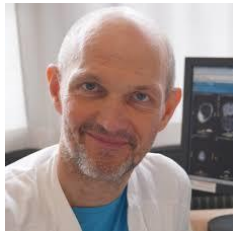
- 22 healthy volunteers
- Four acquisitions for each sequence of the clinical protocol (six for MPRAGE)
- Periodic nodding / shaking motion for 40 seconds during acquisition of k-space centre
- Reacquisition of slices with large amount of motion

TO BE RELEASED ASAP
https://github.com/bids-standard/bids-specification/issues/1171



Example for MPRAGE acquisition

MoCo project team & Funding



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Novo Nordisk and Novozymes Talent Program

For good collaboration we'd like to thank

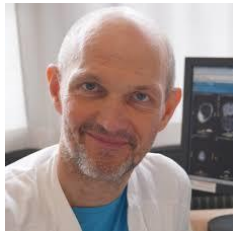
Siemens Healthineers, Denmark
TracInnovations A

https://github.com/melanieganz/MoCoProject

Thank you

Questions?

MoCo project team & Funding



Rigshospitalet

Et højt specialiseret hospital

DET NATUR- OG BIOVIDENSKABELIGE FAKULTET KØBENHAVNS UNIVERSITET

MGH/HST Athinoula A. Martinos Center for Biomedical Imaging

MASSACHUSETTS GENERAL HOSPITAL

METROPOLITAN UNIVERSITY COLLEGE

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